Tzu-Hsuan Chao

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Objective

Seeking an engineer position in computational chemistry, protein design, and drug discovery, leveraging expertise in data-driven modeling and molecular simulations

Education

Texas A&M University, College Station, Texas — Expected June 2026
PhD in Chemistry

National Cheng Kung University, Taiwan — July 2020
MS in chemistry

National Taiwan University, Taiwan — July 2018
BS in Plant Pathology and Microbiology

Experience

Texas A&M University, College Station, TX — Present – August 2021
Research Assistant

- Conducted protein research with data driven approaches and physical-informed feature engineering
- Performed statistical analysis of plastic crystal materials coupled with molecular dynamics to investigate physical properties
- Developed computational tools using machine-learned interatomic potential and transition state finding algorithms to predict mechano-accessible organic reactions

National Cheng Kung University, Tainan, Taiwan — August 2020 – July 2021
Research Assistant

- Investigated reaction mechanisms in surface chemistry and organic chemistry
- Developed theoretical Raman spectrum workflow for metal surfaces and catalytic systems

Activities

Taiwanese Student Association — May 2023 – May 2026
Network Manager (2023-2024)

- Managed association website platforms and supported event coordination and member communications

Fundraising Manager (2024-2025)

- Secured sponsorships from external organizations and strengthened engagement with alumni

Treasurer (2025-2026)

- Prepared financial statements and managed budgets for ongoing and future events

Service

Chemistry Open House — October 2023, 2024, 2025

- Showcased photochemistry to local high school students and community visitors

Skills

Scientific Expertise: Physical Chemistry, Protein Dynamics, Protein Design, Polymer Physics, Coarse-Grained Simulation
Programming Skills: Python, R, C
Software: VASP, Gaussian, Q-Chem, ADF, LAMMPS, GROMACS, cp2k, ORCA, Schrödinger

Honors and Awards

- **2025** Phil Gramm Senator Award
- **2022** NATEA-Dallas Scholarship, North America Taiwanese Engineering & Science Association
- **2020** Gold Award, "English oral competition, 38th Taiwan Symposium of Catalysis and Reaction Engineering
- **2020** Elite Student Scholarship, Elite-Well Education Foundation, Taiwan

Presentations

Poster Presentations

- Predicting Disordered Protein Polymer Physics Directly from Primary Sequence and New Sequence Design, American Chemical Society Spring Conference. **March 2023**. Indianapolis, IN
- Folding Behavior Directly from Sequence, Conference on Advances in Data Science: Theory, Methods and Computation. **October 2022**. College Station, TX

Publications

First Author Papers

- **Chao, T.-H.**; Foncerrada, A.; Shamberger, P. J.; Tabor, D. P. Low-Data Machine Learning Models for Predicting Thermodynamic Properties of Solid-Solid Phase Transformations in Plastic Crystals. *Soft Matter* **2025**, 21, 5957-5968
- **Chao, T.-H.**; Rekhi, S.; Mittal, J.; Tabor, D. P. Data-Driven Models for Predicting Intrinsically Disordered Protein Polymer Physics Directly from Composition or Sequence. *Mol. Syst. Des. Eng.* **2023**, 8, 1146-1155
- Chen, M.-F.[‡]; **Chao, T.-H.**[‡] Shen, M.-H.; Lu, Q.; Cheng, M.-J. Evaluating Potential Catalytic Active Sites on Nitrogen-Doped Graphene for the Oxygen Reduction Reaction: An Approach Based on Constant Electrode Potential Density Functional Theory Calculations. *J. Phys. Chem. C* **2020**, 124(47), 25675-25685. [‡]equal contribution

Selected Co-Author Papers Listed, more in <u>Google Scholar</u>

- Somodi, C.; Satheesh, V.; Chao, T.-H.; Tabor, D. P.; Pentzer, E. B.; Shamberger, P. J. Deuteration of Neopentyl Glycol and Its Effect on the Plastic Crystal Transformation. *J. Phys. Chem. B*. **2025.** Advance article
- He, M.; Chang, X.; **Chao, T.-H.**; Li, C.; Goddard, W. A.; Cheng, M.-J.; Xu, B.; Lu, Q. Selective Enhancement of Methane Formation in Electrochemical CO2 Reduction Enabled by a Raman-Inactive Oxygen-Containing Species on Cu. *ACS Catal*. **2022**, 6036-6046
- Kardile, R. D.; **Chao, T.-H.**; Cheng, M.-J.; Liu, R.-S. Gold(I)-Catalyzed Highly Diastereo- and Enantioselective Cyclization–[4+3] Annulation Cascades between 2-(1-Alkynyl)-2-Alken-1-Ones and Anthranils. *Angew. Chem. Int.* **2020**, 59(26), 10396–10400